

Aditya Tayal

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Final-year CSE undergraduate at IIT Mandi who ships production software end to end. Solo-built PlanMyDegree, now used by **1,741 IIT Mandi students**, plus faculty-leave, exam and lab systems for institute offices. Four startup internships across travel-tech payments, solar-energy ML, counter-drone systems and hiring software; **All India Rank 1** in the SAND 2025 (ICASSP) challenge; elected **UG Academic Affairs Secretary** for ~3,500 undergraduates.

EDUCATION

Indian Institute of Technology Mandi

B.Tech in Computer Science & Engineering | CGPA: 8.80

Mandi, Himachal Pradesh

Aug 2023 – May 2027 (expected)

Relevant Coursework: Machine Learning, Deep Learning (CS-671), Advanced Machine Learning / Generative AI (CS-683), Data Science I & II, Data Handling & Visualization, Database Systems, Design of Algorithms, Automata Theory, Mathematical Foundations of CS, Operating Systems, Computer Networks, Information Security & Privacy, Optimization, Paradigms of Programming

Ramkrishna Mission

Higher Secondary Education in PCM with Computer Science | 89.40%

Gwalior, Madhya Pradesh

Aug 2020 – Jun 2022

Wendy School Junior College

ICSE Secondary Education | 98.20%

Gwalior, Madhya Pradesh

Apr 2007 – Jul 2020

EXPERIENCE

Major Technical Project (MTP), American Express

2026 – 2027 (in progress)

- Final-year Major Technical Project with American Express.

Software Developer Intern, Koha Consultants (InstaFarms & MagoStays)

Apr 2026 – Present

- Full-stack engineer on a two-brand villa & farmstay booking platform (TypeScript backend, admin panel, two customer sites, host mobile app): **490+ commits**, **~170 API endpoints**, **52 SQL migrations** and **100+ test files (~900 cases)** across 5 repositories.
- **Invoice engine:** per-financial-year serial numbers from row-locked counters behind a forward-only trigger; exactly-once issuance via issuance keys and an append-only event trail; maker-checker corrections that issue credit/debit notes or reissues in one transaction.
- **Finance Correction Mode:** a no-write dry run classifies every affected booking, then per-booking reverse-and-replay sub-transactions apply it; closed periods receive append-only deltas instead of rewrites.
- **Tax & accounting:** GST, TCS and TDS across **4 owner agreement models**, verified by a **128-case parameterised test**; month-end owner charges idempotent down to DB CHECK constraints; a transactional outbox to the accounting system with hash-bound idempotency keys and selective retries.
- **Payouts:** Razorpay Route & RazorpayX behind stub/live client seams; payout rows persisted before the network call as idempotency keys, webhooks ordered by event time, a reconciliation sweep and guarded retries; a host wallet with a lock serialising payouts and withdrawals; penny-drop bank verification and KYC.
- **Bookings:** OTA booking & cancellation with a no-write preview, atomic cancel and audited overrides; a per-room Resort property type across all 5 apps that unified three availability implementations and closed **24 catalogued edge cases**.
- **Partner inventory:** hourly catalogue sync with soft deletes, pay-then-book with HMAC verification and an atomic processing state, a retry state machine with refund fallback; cut availability-calendar API calls from **60 to 2**.
- **Apps & quality:** React Native/Expo host app (wallet, payout onboarding, room picker); a plain-language Invoice Test Lab for non-technical testers; admin TypeScript errors **62 → 0**; new SQL validated in rolled-back transactions on staging, plus Playwright end-to-end runs.

AI/ML Engineer Intern, WattCycle (Wattcycle Sustainable Solutions)

Dec 2025 – Jun 2026

- Worked across ML and full stack on WattCycle Atlas, a solar-plant monitoring platform: **17 PRs**; author of the **5,900-line** Python ML service (FastAPI, 14 modules), wired to the Node API through a typed proxy.
- **Fault detection:** 5 rule-based plant-health checks with severity tiers plus an **Isolation Forest** detector over 12 features; 24-hour fault de-duplication; hourly checks, 6-hourly anomaly scans and weekly retraining schedules.
- **Forecasting:** a **13-horizon** forecaster choosing XGBoost (1–24 h), a Prophet + XGBoost ensemble (2–7 days) or Prophet (up to a year) on 34 engineered features with MAE/RMSE/MAPE/R² tracking; LSTM and MLflow tracking prepared for rollout.
- **Dashboard backend:** TimescaleDB aggregation tables, 2 concurrently refreshed materialised views, PL/pgSQL functions, **7 role-protected endpoints**, 3 cron jobs and idempotent upserts.
- **Data integrity & platform:** replaced random chart placeholders with sourced values and a 15-category metrics spec; pnpm monorepo with one lockfile, Dockerised services, CI caching, Dependabot; **87 tests** behind an 80% coverage gate.

Full Stack Developer Intern, Garudakshak Technologies (Counter-UAV Defence)

Oct 2025 – Apr 2026

- Created and solely developed the operator console for a counter-drone sensing system, from an empty repository to **68 releases** for Windows and Raspberry Pi.

- Tauri v2 desktop app whose Rust shell health-checks and launches a Go API and a Python ML service behind a React + TypeScript UI.
- Grew the Go API gateway from 8 to **91 routes**, with 10 PostgreSQL/TimescaleDB migrations, Swagger, per-IP rate limiting, security headers, Prometheus metrics, Redis caching and graceful shutdown.
- Keycloak SSO with a local-JWT fallback, token revocation and capability-based permissions on **72 routes**; a WebSocket hub streaming detection, incident and device events to live dashboards and maps.
- A capture watchdog that restarts stalled camera streams with backoff, camera hot-swap and systemd-supervised services on Raspberry Pi 5; integrated the team's ONNX detection models; CI-built MSI/EXE and ARM64 **.deb** releases.

Lead Developer Intern, Strive Partners (Strive Hunt)

Sep 2025 – Jun 2026

- Lead developer of a hiring platform with candidate and employer portals and an internal applicant-tracking system (ATS): **1,000+ commits**, 39 pages, plus a React + Vite + TypeScript public site and candidate dashboard.
- ATS with role-based access, a spreadsheet-style candidate grid (autosave, status pipeline, custom columns), Excel bulk import with column mapping, soft delete with restore, and an audit log.
- Real-time collaboration over WebSockets with backoff reconnects and heartbeats; candidate search with **16 filters** and boolean queries; an interview scheduler with a calendar view.
- One API interceptor for JWT auto-refresh, timeouts, retries and CSRF; inactivity timeouts, XSS sanitising, Google sign-in and email OTP.
- Integrated the Django REST API on AWS (EC2, HTTPS); shipped on Vercel and as a hardened Docker/nginx image with CI container smoke tests and Trivy vulnerability scans.

PROJECTS

Software for IIT Mandi

PlanMyDegree: Academic Planner for IIT Mandi (Solo)

Next.js 15, React 19, TypeScript, Prisma, PostgreSQL, NextAuth, Vercel, PWA

Jan 2026 – Present

- Solo-built a full-stack academic planner now used by **1,741 IIT Mandi students**; **150+ sign-ups in 36 hours** from one WhatsApp message, no ads.
- Tracks **17,555 enrollments** and **51,915 credits**; students built **347 pre-registration plans**; covers 1,427 courses, 13 branch curricula, 15 minors and 5 German exchange universities.
- Pre-registration with seat caps, mutual exclusion of equivalent courses and declarations synced to Google Sheets via an authenticated webhook; transcript OCR import, timetable clash detection, calendar export.
- **30 data models**, **~65 API routes**, **880+ commits**; average feedback rating 4.9 from 27 reviews.

Faculty Leave Portal: Dean of Faculty Affairs Office (Solo)

Pilot

Fastify, Prisma, React, Vite, Tailwind, SQLite, Docker, Raspberry Pi

Sep 2026 – Present

- Built Phase 1 of faculty leave management from the institute's PRD: rule-checked applications, an auditable approval chain and establishment records; traced **324 PRD requirements** with **321 automated tests**.
- No stored balances: leave is derived on every read from an append-only ledger; DB triggers block edits on ledger and audit tables; two-phase HOLD → DEBIT posting; script-hashed OTP, CSRF, magic-byte upload checks.

TestPlatform: Secure Online Examination System (Solo)

Next.js, Express, Socket.IO, PostgreSQL, Redis, nginx, pm2

Aug 2026 – Present

- Student, teacher and admin portals with multi-tenant organisations; server-authoritative exam timers in Redis with Socket.IO live rooms; auto-grading for MCQ, multi-select and numeric-with-tolerance answers.
- Passwordless OTP over email and WhatsApp; hardened after a self-audit: pinned JWT algorithm, row-locked final submit (SELECT ... FOR UPDATE), IDOR fixes, 2 CVE patches.

Exodia 2026 Ticketing and Gate Management System (Solo)

Code

Next.js 15, tRPC, Prisma, Neon Postgres, NextAuth, Nodemailer

Mar 2026 – Apr 2026

- Built QR ticketing, email delivery and gate scanning for the fest's Pronite: **300+ QR tickets** emailed, **200 attendees** admitted, **0 duplicate entries**.
- Atomic single-use QR claims so concurrent gate scans cannot double-admit; **blocked 26 reuse attempts**; shipped **13 commits live** on event day with zero downtime.

Faculty Application Portal, IIT Mandi

Live

Django, SQL, HTML/CSS

Dec 2024 – Feb 2025

- Replaced Google Forms with a secure Django portal processing **200+ faculty applications** annually; admin dashboard to filter, track and export; automated CV generation & emails, saving **20+ admin hours/month**.

Atal Tinkering Lab Tools: TinkerBench, Simulation Lab, NCERT Sim (Solo)

JavaScript, Canvas, nginx, Docker

Sep 2026 – Oct 2026

- Browser electronics bench solving circuits by nodal analysis every frame (diodes, transistors, a working 555 timer) and running Arduino C++ translated to JS; **23 Class 9–12 experiments** and **26 NCERT physics practicals**.

IIT Mandi LaTeX: Institute-hosted Overleaf Alternative (Lead)

Overleaf CE, Node.js, Docker, CI

Sep 2026 – Present

- Extended Overleaf CE for ~3,500 users: FCFS compile queue, 50-user cap with waiting room, 500 MB quotas, nightly off-host backups; costed at **INR 1.26 lakh one-time vs INR 11.29 lakh/yr**.

Machine Learning & Research

Camouflaged Object Detection (Paid Research Project)

PyTorch, mmdetection, DINOv2, LoRA, SigLIP-2, SAM2

Apr 2026 – Jul 2026

- Adapted PAIS (ICCV 2023) to camouflaged objects with 10% labels: COD10K AP **29.7** → **33.0**, **+7.8 AP** over the supervised baseline; NC4K AP 46.9.
- Designed LGCAD, a DINOv2 + LoRA decoder with SigLIP-2 text-guided queries (22 variants, 4 datasets): S-measure **0.847** (COD10K), **0.865** (NC4K).
- Reproduced training-free UCOD-DPL (CVPR 2025) with two-pass SAM2 refinement and 25-mask soft voting: S-measure **0.8407 vs paper's 0.835**; improved GBNet on **42/42** dataset-metric checks over 71 experiment phases.

Domain Unlearning in CLIP (CS-683, Team of 4)

[Code](#)

PyTorch, CLIP ViT-B/16, Prompt Learning, SCRUB

Jan 2026 – May 2026

- Re-implemented ADU and beat the published Office-Home H-score (**69.96** → **72.10**) by fixing an N:1 loss imbalance between kept and forgotten domains.
- Adapted SCRUB from sample- to domain-level unlearning: **83.6 H-score**, **+13.6 points** over the paper; 82.9 average across 6 two-domain pairs; IEEE-format report.

Vehicle Counting with Classical CV (Vehant Challenge, Solo)

OpenCV, NumPy

Jan 2026

- 9-stage pipeline in 48 hours with deep learning banned (GMM background subtraction, motion ROI, centroid tracking, track merging, gate-line counting, night mode): **MAE 16.4**, **rank 10** against teams.

MahaDrishti AI: Satellite Crop Verification (Pune Agri Hackathon Finalist)

Sentinel-1/2, Prithvi-EO 2.0, Random Forest, FastAPI, Gemini

May 2026

- Verifies farmers' declared crops per land-record plot from satellite imagery with Marathi/English reports; RF 5-fold CV **81.86%**, mixture-of-experts **83.21%**; 194 plots per village in under 2 minutes.

Product Price Prediction (Amazon ML Challenge 2025, Team of 3)

LightGBM, TF-IDF, Python

Oct 2025

- TF-IDF + 18 regex features into a 21-model LightGBM ensemble (7-fold × 3 seeds) with calibration: SMAPE **54.38%** → **32.33%** out-of-fold.

Deepfake Detection using Vision Transformers (CS-671, Team of 6)

CLIP ViT-L/14, ViT-H/14, DINOv2, SVM (cuML)

Feb 2025 – May 2025

- Frozen ViT features + GPU RBF-SVM evaluated on **19 generators** and **80,353** test images: **89.4%** accuracy weighted by test-set size.

CaptureSmart: AI Camera Assistant for Android (Team of 6)

[Code](#)

Kotlin, CameraX, TensorFlow, MobileNetV2, Flask

May 2025

- MobileNetV2 blur scorer (MAE 6.7, Pearson 0.89 on 24,000 images) plus ISO/shutter recommenders (shutter R² 0.912); secured **3rd** place among **60+** teams.

Applied AI & Hackathon Builds

FinFabric: Applied AI for Banking (IDBI Innovate 2026, Team Lead)

[Code](#)

JavaScript, Vercel

Jul 2026

- Led a 5-product live suite: MSME health card, mutual-fund SIP companion with live NAVs, lead scoring, default early-warning model and a fuzzy-matching reconciliation copilot.

Sahayak: Conversational AI for a Crime Database (KSP Datathon 2026)

[Code](#)

Qwen 2.5 Coder 7B, Zoho Catalyst, FastAPI, React

Jun 2026

- English/Kannada questions to SQL over 3,500 FIRs with a read-only SQL guard; every answer shows its SQL and source rows; IPC ↔ BNS legal-code crosswalk.

SevaVoice (HACKHAZARDS '26) and DeepMatch (India Runs)

[Live Code](#)

Sarvam AI, React, TypeScript, MiniLM, Python

Jun 2026

- SevaVoice turns public-service notices into verified action plans in **14 Indian languages** (voice in 11), with 63 Playwright checks.
- DeepMatch ranks candidates in 3 stages (rules 100K → 5K, MiniLM scoring, signal fusion) with 6 fake-profile heuristics in ~75 s on CPU.

DyslexoFly: Accessible Reading Platform (Code for Bharat S2)

[Live Code](#)

Next.js 15, Flask, DistilBART, mT5, edge-tts

Jun 2025 – Jul 2025

- Converts PDF/DOCX/TXT into dyslexia-friendly reading with OpenDyslexic, English and Hindi summaries, and audio in 7 neural voices.

Adobe India Hackathon 2025: PDF Intelligence (Team Leader)

[Code](#)

Python, PyMuPDF, MiniLM, Docker

Jul 2025

- Reached Round 2 (**top 1.5%** of 262,120 teams): multilingual PDF outline extraction in an offline CPU-only Docker image and persona-aware ranking of 7 PDFs in 30–45 s.

More builds:

- **FlowAI** (Vibe2Ship 2026): Gemini multi-agent productivity app on Next.js + Firebase. **PurrPedia** (HackTheKitty 2026): Next.js 16 + Temporal workflows; fixed all 5 security-audit findings.
- **TheBudgetBuddy** (FrostHack 2025): LangChain agent with Groq Llama3-70B, Cohere rerank and FAISS. **CareerAutofill AI**: agents + Chrome extension; LoRA on distilgpt2 cut perplexity 35.7 → 23.2.
- **CyberShield** (PSBs Hackathon, IIT Hyderabad): mule-account detection on 9,082 accounts (0.89% base rate). **Vidyut AI** (ET AI Hackathon 2.0): citation-grounded Q&A over safety documents.
- **Face EngagementCV**: dlib + MediaPipe face ID and engagement scoring from CCTV. **PSCDL 2026**: classical-CV scene change detection. **TheAttendioBot**: Telegram attendance planner.
- **SAND 2025**: ALS severity from voice (macro-F1 0.5430). **ISTP**: solar PV waste policy study with field visit. **Autonomous ROS robot**: ESP32 on a custom 2-layer PCB.
- Also: malware scanner (FastAPI + YARA), SecureInvest-AI (SEBI GFF 2025), PodSense (ABB 2026), UGAC website, GCS website, Neufin, PRAYAS ML labs, LaTeX lecture notes.

POSITIONS OF RESPONSIBILITY

- UG Academic Affairs Secretary, Student Gymkhana, IIT Mandi** *Jul 2026 – Present*
- Elected student lead for academics for ~**3,500 undergraduates**; head the UG Academic Council; selected and oriented **12 branch councillors**.
 - Ran Batch 2026 academic orientation and **15 branch-wise sessions**; prepared proposals for the 65th and 66th Board of Academics, including a **388-page** curriculum booklet.
 - Proposed an institute-hosted LaTeX platform backed by a **483-response** survey; audited the Academic Section website with **34 changes** adding 568 approved courses.
 - Organised Teachers' Day 2026 (chief guest Prof. Prem Kumar Kalra), reaching more than 60% of faculty.
- Batch 2023 Academic Representative & Academic Guide, UG Academic Council** *2024 – 2026*
- Liaised between **500+ students** & faculty; co-presented the first post-COVID offline Curriculum Explanation Session; hired the UGAC web team (8 candidates, 4 selected); organised *VyarthShastra*.
- Head of Technical Team, Exodia 2026 (Annual Cultural Fest)** *Apr 2026*
- Owned fest tech: solo-built the ticketing and gate system and tuned exodia26.org performance.
- Instructor, HIMSHIKHAR (CCE IIT Mandi × Masai) & PRAYAS 4.0** *Jun 2026 – Jul 2026*
- Taught **22 classes (40+ hours)**, DBMS to GenAI, and an 18-hour ML course with 4 Colab labs; led **8 TAs**.
- UG Head Teaching Assistant, Machine Learning**
- Supervised course operations for **600+ students** across a full semester.
- Mentoring & Clubs** *2023 – Present*
- GCS Cell Mentor; SAE Club Mentor; Core Member of STAC (hackathons for **100+ teams**); JEE Mentor at Physics Wallah (**100+ aspirants**).

ACHIEVEMENTS

- AIR 1, Global Rank 11, SAND 2025 (IEEE ICASSP 2026 challenge)**: ALS severity from voice; India #2, Global 19 in Task 2.
- Rank 10, solo vs teams, Vehant Vehicle Count Challenge**: classical CV only, MAE 16.4 (Jan 2026).
- Finalist, Pune International Agri Hackathon 2026**: Geospatial Crop Survey track (May 2026).
- 3rd Rank, AiHack India Hackathon 2025**: hosted at IIT Mandi by Aiful Corporation, Japan (Nov 2025).
- Round 2 (top 1.5%), Adobe India Hackathon 2025**: team leader among 262,120 teams (Jul 2025).
- Emerald of Gymkhana Award**: best GCS Cell volunteer, Dean Students' Office (May 2025).
- National Semi-Finalist, Tata Imagination Challenge 2024**: through 2 national rounds (Feb 2025).
- 1st Rank, Space Code Challenge 2023**: STAC, IIT Mandi (Jan 2024).
- 20+ hackathons & data challenges in 2025–26**, incl. Amazon ML Challenge 2025 (82,787 teams), IDBI Innovate 2026 (team lead), ABB Accelerator 2026, SEBI GFF 2025.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, C, C++, Java, Kotlin, SQL, Rust (Tauri), R, HTML/CSS, LaTeX
Backend: FastAPI, Flask, Django, Node.js, Express, Fastify, Hono, tRPC, REST, OpenAPI, WebSockets, Socket.IO, Kafka
Frontend & Apps: React, Next.js, Vite, Tailwind, Astro, Tauri, React Native (Expo), Android (Kotlin), PWAs
Data: PostgreSQL, SQLite, MongoDB, Redis, Prisma, Drizzle, Firestore, FAISS
ML & AI: PyTorch, TensorFlow/Keras, scikit-learn, LightGBM, XGBoost, OpenCV, Transformers, LoRA/PEFT, CLIP, DINOv2, SAM2, LangChain, RAG, MLflow
Cloud & DevOps: Docker, AWS (EC2, RDS, S3), DigitalOcean, Vercel, Render, Firebase, nginx, pm2, Caddy, CI/CD, Playwright, Linux, Raspberry Pi
Integrations: Razorpay (Route, RazorpayX), Zoho Books, WhatsApp Cloud API, Keycloak SSO, NextAuth/OAuth